I am working on automating a workflow involving FreeCAD to import .csv files, create geometry, and export .step files. These .step files are used in Solid Edge (Siemens) and Patran for FEA analysis. While this workflow worked before, I am now encountering issues. My goal is to develop a robust Python script for FreeCAD to handle this process correctly. Key Details and Context General Workflow: Import .csv files with point coordinates and line elements into FreeCAD. Generate geometry (points, lines, and optionally surfaces) in FreeCAD. Export the geometry as .step files for use in FEA workflows. Issues Encountered: Points are not correctly plotted in FreeCAD: Points accumulate at the edges of lines or are missing entirely. Exported .step files do not retain the intended geometry, causing issues when imported into Solid Edge or Patran. The problem seems to stem from the FreeCAD geometry creation or export process. Sample .csv Files: Below are examples of the .csv files I use: Points (points.csv): x,y,z
31.6357177516464,18.1323048323686,0.01
30.7592787611228,17.6271636565544,0.01
29.8828397705992,17.1220224807403,0.01
29.0064007800756,16.6168813049261,0.01
28.129961789552,16.111740129112,0.01
27.2535227990283,15.6065989532978,0.01
26.3770838085047,15.1014577774836,0.01
25.5006448179811,14.5963166016695,0.01
24.6242058274575,14.0911754258553,0.01
23.7477668369339,13.5860342500412,0.01
22.8713278464103,13.080893074227,0.01 Lines (lines.csv): Copy Edit
Point1,Point2 1,2 2,3 3,4 4,1 Export Requirements: The .step files should adhere to proper tolerances and use schema AP214. They must be compatible with Solid Edge and Patran. Python Script Overview Here is my current FreeCAD script for importing .csv files and exporting .step files: python Copy Edit import FreeCAD as App import Part import csv # Function to read points from a .csv file def read_points(file_path): points = [] with open(file_path, 'r') as csvfile: reader = csv.DictReader(csvfile) for row in reader: x = float(row['x']) y = float(row['y']) z = float(row['z']) points.append(App.Vector(x, y, z)) return points # Function to read elements from a .csv file def read_elements(file_path): elements = [] with open(file_path, 'r') as csvfile: reader = csv.DictReader(csvfile) for row in reader: elements.append((int(row['Point1']) - 1, int(row['Point2']) - 1)) return elements # Create geometry in FreeCAD def create_geometry(points, elements, doc): for i, (p1_index, p2_index) in enumerate(elements): line = Part.LineSegment(points[p1_index], points[p2_index]) edge = line.toShape() obj = doc.addObject("Part::Feature", f"Edge_{i}") obj.Shape = edge doc.recompute() # Export geometry as STEP file def export_to_step(doc, output_path): App.ParamGet("User

parameter:BaseApp/Preferences/Mod/Part").SetInt("STEPExportSchema", 214) App.ParamGet("User parameter:BaseApp/Preferences/Mod/Part").SetFloat("STEPExportPrecision", 1e-6) objects = [obj for obj in doc.Objects if obj.TypeId.startswith("Part::")] Part.export(objects, output_path) print(f"Model exported as STEP to: {output_path}") # Main script def main(): points_csv = "points.csv" elements_csv = "lines.csv" step_output = "output_model.step" points = read_points(points_csv) elements = read_elements(elements_csv) doc = App.newDocument("GeometryModel") create_geometry(points, elements, doc) export_to_step(doc, step_output) if __name__ == "__main__": main() Questions and Goals Geometry Creation: How can I ensure all points are correctly incorporated into the geometry, so they are retained during export? Is there a way to visualize points in FreeCAD that are not part of any geometry for debugging? STEP File Export: What are the best practices for exporting .step files in FreeCAD for FEA workflows? Are there specific export settings (e.g., schema, tolerances) to ensure compatibility with Solid Edge and Patran? Debugging Suggestions: How can I validate .csv files before importing them into FreeCAD to avoid geometry issues? What checks can I add to my script to handle missing points or invalid line connections? Script Optimization: Are there improvements or best practices you would recommend to make the script more robust and efficient?

ChatGPT

Thought about FreeCAD automation workflow for 16 seconds

Below is a collection of suggestions, best practices, and debugging tips to help you refine your FreeCAD workflow for importing CSV data, creating geometry, and exporting STEP files that remain compatible with Solid Edge and Patran. Some of these points address the geometry issues you mentioned (e.g., missing or misplaced points, edges not recognized properly), while others focus on improving robustness and maintainability of your script.

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1. Ensuring Points are Properly Incorporated into the Geometry

Why Are Points Sometimes Missing in STEP?

- By default, STEP files do not necessarily preserve “standalone” points (0D geometry) unless they are incorporated into a recognized geometry entity (e.g., vertices in an edge or curve, or annotation objects in some CAD systems).

- In your script, you create Part.LineSegment objects from pairs of points. The lines themselves will appear as edges in the STEP file, but the original points may not be exported as separate geometrical entities.

How to Retain Points If Needed

1. **Create Vertex Shapes**:
   If you truly need those points recognized as geometry (for example, for meshing references in FEA), you can create vertices:

python

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```
# Create a Part Vertex for each point

for i, pt in enumerate(points):

  vertex_shape = Part.Vertex(pt)

  vertex_obj = doc.addObject("Part::Feature", f"Vertex_{i}")

  vertex_obj.Shape = vertex_shape
```

Note: Many FEA tools focus more on edges/surfaces/volumes than standalone vertices. If you only need them for debugging or reference in FreeCAD, using a "Draft Point" object might suffice.

2. **Use Draft Workbench Tools** (Optional):
   If you prefer the Draft workbench approach (instead of pure Part geometry), you can use:

python

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```
import Draft

Draft.makePoint(pt[0], pt[1], pt[2])
```

These points will be visible in FreeCAD's 3D view but may not always export as geometry to a STEP file.

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2. Visualizing (Standalone) Points for Debugging in FreeCAD

1. **Draft Points**:
   As mentioned, Draft.makePoint(...) is an easy way to quickly visualize a set of points in FreeCAD's 3D view.

2. **Set Display Preferences**:
    - In FreeCAD, go to Edit > Preferences > Display > Points and adjust point size, color, etc., to better spot them in the viewport.
    - Check the View tab of each object (in the Combo View) to ensure "Visibility" is on.

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3. Creating Edges, Wires, or Surfaces

Combine Edges into a Single Object

- Having every line segment as its own Part::Feature object can be cumbersome (especially if you have hundreds or thousands of lines).
- You can combine all edges into a single Part::Compound or create wires if the lines form continuous loops:

python

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```
import Part


edges = []
for i, (p1_index, p2_index) in enumerate(elements):
  line = Part.LineSegment(points[p1_index], points[p2_index])
  edges.append(line.toShape())


# Combine all edges into one compound
compound = Part.makeCompound(edges)
compound_obj = doc.addObject("Part::Feature", "AllEdges")
compound_obj.Shape = compound
doc.recompute()
```

Creating a Wire (if lines form a loop)

- If the lines form one or more closed loops, you can create wires or faces from them for more "solid-like" geometry that is often friendlier for FEA:

python

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```
wire = Part.Wire(edges)    # if they form a continuous chain
face = Part.Face(wire)     # optional, if planar
```

- Surfaces or solids are typically more useful for FEA meshing than just line segments.

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4. STEP Export Best Practices for FEA Workflows

1. STEP Schema and Tolerances

- **Schema**: You are already setting AP214, which is generally well-accepted (also for Solid Edge and many CAE tools). AP203 could be used as well, but AP214 typically includes a richer data structure for colors/layers.
- **Tolerances**: You set STEPExportPrecision to 1e-6. For many mechanical/FEA contexts, a tolerance in the 1e-6 to 1e-7 range is typical. If you see geometry or meshing issues (e.g., free edges in a mesh when you think they should be continuous), sometimes reducing tolerance further may help, but it can also bloat file sizes or slow down export. It is a balancing act.

2. Ensuring Geometry Is Recognized

- **Single Unified Shape**: FEA and CAD software often prefer a single "solid" or a well-defined surface entity. If you only have edges, you may want to confirm that Solid Edge or Patran interpret those edges in the intended way (e.g., as curves for boundary conditions, beam elements, or something else).
- **Check for Null Shapes**: Before exporting, validate that each shape is valid:

python

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```
for obj in doc.Objects:
  if hasattr(obj, "Shape"):
    if not obj.Shape.isValid():
      print(f"Invalid shape detected in {obj.Name}")
```

- **Set Proper Names**: Some software is picky about naming/labeling. You can rename the compound or feature to something meaningful (e.g., "WireModel") if needed.

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5. Debugging and Validation

1. Validate the CSV Files

- **Check Number of Columns**: Ensure each row has the required columns x, y, z (for points) or Point1, Point2 (for lines). Handle errors gracefully:

python

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```
try:
  x = float(row['x'])
  ...
except (KeyError, ValueError) as e:
  # Log or raise an error
```

- **Check Index Range**: When reading lines, confirm that Point1 and Point2 indices do not exceed the length of the points list. You can skip or log invalid references.

2. Add Sanity Checks

- **Distance Checks**: If you know expected distances, you can quickly detect if points are all at zero or extremely large coordinates.
- **No Duplicate Lines**: Optionally check if (p1_index, p2_index) repeats, which could cause unexpected geometry overlaps.

3. Use the "Part Check Geometry" Tool in FreeCAD

- In the FreeCAD GUI, after generating your geometry, you can select your object in the Tree View and go to Part → Check geometry. This can help diagnose geometric errors that sometimes lead to poor exports.

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6. Script Optimization and Robustness

1. **Create One Document Object vs. Many**
   Instead of creating a separate Part::Feature for each line, build up the

geometry in Python and then create **one** object in the document. This is much more efficient and keeps the FreeCAD tree simpler.

2. **Use a Context Manager or Cleanup**
   If you're running FreeCAD in headless mode repeatedly, be sure to close or clean up documents after export to avoid memory bloat:

```
python

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doc = App.newDocument("GeometryModel")

try:

  # your geometry creation and export

finally:

  App.closeDocument(doc.Name)
```

3. **Error Handling**
   - Wrap read operations in try/except blocks.
   - Log errors (e.g., to a text file or standard output) if an element references a non-existent point.
4. **Performance**
   - If you have thousands of edges, you may benefit from building them into a Part.Compound or a single object as described.
   - Only call doc.recompute() once after creating all geometry, rather than after each edge.

Putting It All Together: A More Robust Example

Below is a modified version of your script illustrating some of these ideas. Note the single Part::Compound creation and optional vertex creation. Adjust to your needs:

```
python

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import FreeCAD as App

import Part

import csv
```

```
def read_points(file_path):
    """Reads CSV with columns 'x', 'y', 'z'."""
    points = []
    with open(file_path, 'r') as csvfile:
        reader = csv.DictReader(csvfile)
        for row in reader:
            try:
                x = float(row['x'])
                y = float(row['y'])
                z = float(row['z'])
                points.append(App.Vector(x, y, z))
            except (KeyError, ValueError):
                print(f"Skipping invalid point row: {row}")
    return points

def read_elements(file_path):
    """Reads CSV with columns 'Point1', 'Point2'."""
    elements = []
    with open(file_path, 'r') as csvfile:
        reader = csv.DictReader(csvfile)
        for row in reader:
            try:
                p1 = int(row['Point1']) - 1
                p2 = int(row['Point2']) - 1
                elements.append((p1, p2))
            except (KeyError, ValueError):
                print(f"Skipping invalid line row: {row}")
```

```
    return elements

def create_geometry(points, elements, doc, create_vertices=False):
    """Create edges from point indices. Optionally create separate vertices."""
    edges = []
    n_points = len(points)

    # Create edges
    for i, (p1_index, p2_index) in enumerate(elements):
        if p1_index < 0 or p1_index >= n_points or p2_index < 0 or p2_index >= n_points:
            print(f"Warning: Invalid element {i} references out-of-range index.")
            continue
        line = Part.LineSegment(points[p1_index], points[p2_index])
        edges.append(line.toShape())

    # Combine edges into a single compound
    if edges:
        compound = Part.makeCompound(edges)
        compound_obj = doc.addObject("Part::Feature", "AllEdges")
        compound_obj.Shape = compound

    # Optionally create vertices (for debugging or specific workflows)
    if create_vertices:
        for i, pt in enumerate(points):
            vtx_shape = Part.Vertex(pt)
            vtx_obj = doc.addObject("Part::Feature", f"Vertex_{i}")
            vtx_obj.Shape = vtx_shape
```

```
    doc.recompute()


def export_to_step(doc, output_path):
    # Set STEP export parameters
    App.ParamGet("User parameter:BaseApp/Preferences/Mod/Part").SetInt("STEPExportSchema", 214)
    App.ParamGet("User parameter:BaseApp/Preferences/Mod/Part").SetFloat("STEPExportPrecision", 1e-6)

    # Gather all Part objects
    objects = [obj for obj in doc.Objects if obj.TypeId.startswith("Part::")]
    if not objects:
        print("No Part objects found to export.")
        return

    # Export
    Part.export(objects, output_path)
    print(f"Model exported as STEP to: {output_path}")

def main():
    points_csv = "points.csv"
    elements_csv = "lines.csv"
    step_output = "output_model.step"

    points = read_points(points_csv)
    elements = read_elements(elements_csv)

    doc = App.newDocument("GeometryModel")
```

```
    create_geometry(points, elements, doc, create_vertices=True)  # toggle as needed
    export_to_step(doc, step_output)

    # Optional: close the document when done if running headless
    # App.closeDocument(doc.Name)

if __name__ == "__main__":
    main()
```

Explanation of Key Changes:

- **create_vertices flag**: allows you to decide if you want separate vertices in your STEP file (or for debugging in FreeCAD).
- **Single Part::Feature**: consolidates all edges into one object for cleanliness and easier handling in external FEA tools.
- **Error/Range Checks**: prints warnings if a line references a non-existent point.
- **Validation**: quick checks for invalid CSV rows.

Summary

1. **Retain Points**: Decide whether you truly need them in the final STEP (as vertices) or just for debugging.
2. **Consolidate Geometry**: Combine edges into a single compound (or wire/face/solid) for better FEA compatibility.
3. **Use Proper STEP Settings**: AP214 with a reasonable tolerance (e.g., 1e-6) is typically fine for Solid Edge and Patran.
4. **Debugging**: Validate CSV data, watch out for out-of-range references, and use FreeCAD's geometry check.
5. **Efficiency & Maintenance**: Minimize the number of objects in the tree and handle potential errors gracefully.